

PROGRAM-5: Demonstrate CI integration by running jenkins in a Docker container

Project Structure:

program-5/

```
|— Dockerfile
|— docker-compose.yml
```

Step 1: Ensure Docker and Docker Compose are Running

```
docker --version
```

```
docker compose version
```

Step 2: Create the Project Directory

```
mkdir program-5
```

```
cd program-5
```

Step 3: Create the Dockerfile

```
Nano Dockerfile
```

```
FROM jenkins/jenkins:lts
```

```
# Switch to root user to install tools
```

```
USER root
```

```
# Install Docker inside Jenkins container
```

```
RUN apt-get update && apt-get install -y docker.io
```

```
# Switch back to Jenkins user
```

```
USER jenkins
```

Step 4: Create docker-compose.yml File

```
version: '3.8'
```

```
services:
```

```
  jenkins:
```

```
    build: .
```

```
    container_name: jenkins-ci
```

```
    ports:
```

```
      - "8080:8080"
```

```
      - "50000:50000"
```

```
    volumes:
```

```
      - jenkins_home:/var/jenkins_home
```

```
      - /var/run/docker.sock:/var/run/docker.sock
```

```
volumes:
```

```
  jenkins_home:
```

Step 5: Build and Run Jenkins Container

```
docker compose up -d
```

```

1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ docker compose version
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ mkdir program-5
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ cd program-5
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-5$ nano Dockerfile
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-5$ nano docker-compose.yml
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-5$ docker compose up --build -d
WARN[0000] /home/chandana-s/program-5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 33.3s (8/8) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 513B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 252B
=> [internal] load metadata for docker.io/jenkins/jenkins:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:latest@sha256:7b1c378278279c8688efd6168c25a1c2723a6bd6f0420beb5ccefabee3cc3bb1
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting to image
=> => exporting layers
=> => writing image sha256:5e5b18259aa0b578967c2a0bd584c1a663cda33acdf8fa68b91cb11e1e934a5
=> => naming to docker.io/library/program-5-jenkins
=> resolving provenance for metadata file
[+] Running 2/2
  ✓ program-5-jenkins      Built
  ✓ Container jenkins-ci    Started
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-5$ docker exec -it jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword
5947292739df4e6ab7bfbcb05ee144768
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program-5$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
program-5-jenkins:latest	5e5b18259aa0	925MB	0B	

Step 6: Access Jenkins Web Interface

<http://localhost:8080>

